

Final Group Project: Dataset Exploration and Initial Analysis

For the final project of this course, each group initially should select a dataset and conduct an exploratory analysis. The goal of this first stage of the project is to understand the structure and characteristics of the dataset.

Guidelines for choosing datasets: the dataset should contain **at least 1000 observations (rows)** and **at least 10 features (columns)**. It is preferred that the dataset include both numerical and categorical variables **if possible**. The dataset should also come from a reliable source (e.g., Kaggle, UCI repository, government open data portals, or similar sources).

In this phase of the project, your objective is to **explore and describe the dataset using the statistical concepts and visualization techniques** we have learnt so far, such as central tendency, dispersion, distribution shape, skewness, and exploratory data analysis (EDA). The emphasis should be on understanding the data and interpreting your findings, not just producing plots or tables.

Your report should clearly document **the following tasks**:

1. Dataset Description

- Source of the dataset and the real-world problem it relates to
- Total number of observations and variables/features
- A brief description of each feature in a table (feature name, type, meaning and classification such as numerical (continuous/discrete) or categorical (nominal/ordinal/binary))

2. Central Tendency and Dispersion Analysis

- Compute and report **mean, median, mode, variance, standard deviation, and IQR (if meaningful)** for numerical features in a table, and discuss what they indicate about the distribution of the features

3. Distribution and Shape

- Visualize the distribution of **numerical** features using **histograms, density plots, and boxplots**
- Comment on the shape of the distribution (e.g., symmetric, right-skewed, left-skewed, heavy-tailed)
- Identify outliers using the IQR rule

4. Categorical Variable Analysis

- Create **frequency tables and bar/pie charts** for categorical variables
- Identify the most common categories and comment on any **imbalance** in the data

5. Relationships Between Variables

- Use **scatterplots** to explore relationships between different pairs of numerical variables (e.g., relationship between *price* and *condition*)

6. Data Quality Assessment

- Identify **missing values, duplicate rows, or unusual values** in the dataset
- Briefly discuss how such issues might affect analysis and how they could be handled

7. Key Insights

- Summarize the **most important patterns or insights** discovered during the analysis
- Mention which variables appear most interesting or potentially useful for **future predictive modeling or analyses**

Note: It is preferable to use tables in the report to present data/values where applicable.